

E-COMMERCE SALES & CUSTOMER ANALYTICS

SQL Project Report

Prepared from Ecommerce_SQL_Project_Complete.xlsx

1. Project Overview

This project analyses an e-commerce dataset using SQL-oriented data modelling and analytical techniques. The objective is to understand customer behaviour, product performance, order activity, revenue contribution and geographic sales patterns. The workbook contains four related tables: customers, products, orders and order_items.

2. Objectives

- Measure total revenue, order volume and average order value.
- Identify the highest-performing products and categories.
- Analyse revenue by month and customer city.
- Find high-value customers and repeat purchasing behaviour.
- Demonstrate joins, aggregation, subqueries, CTEs, CASE expressions and window functions.

3. Dataset Description

Table	Purpose	Important columns	Rows
customers	Customer master data	customer_id, customer_name, city, signup_date	50
products	Product catalogue and prices	product_id, product_name, category, price	20
orders	Order-level transactions	order_id, customer_id, order_date, total_amount	150
order_items	Product lines within each order	order_id, product_id, quantity, unit_price	383

The dataset contains 50 customers, 20 products, 150 orders and 383 order-item records. Dates in the order table span 2024-01-04 to 2024-12-31.

4. Data Model and Relationships

The customers table is connected to orders through customer_id. The orders table is connected to order_items through order_id. Products are connected to order_items through product_id. This structure separates master data from transaction data and supports accurate analysis at both order and product-line level.

Relationship	Join condition	Meaning
customers → orders	customers.customer_id = orders.customer_id	One customer can place many orders.
orders → order_items	orders.order_id = order_items.order_id	One order can contain multiple products.
products → order_items	products.product_id = order_items.product_id	One product can appear in many order lines.

5. Data Preparation and Quality Checks

The workbook is already organised into analysis-ready sheets. Before querying, the following checks should be performed: verify primary-key uniqueness, confirm that foreign keys match parent records, inspect null values, validate numeric datatypes, and compare order totals with the sum of order-item line values. These checks reduce the risk of incorrect joins and misleading totals.

6. SQL Concepts Applied

SQL concept	Application in this project
SELECT / WHERE	Choose fields and filter orders or products.
GROUP BY / aggregate functions	Calculate revenue, quantity, order counts and averages.
HAVING / ORDER BY / LIMIT	Filter grouped results and rank top performers.
INNER JOIN	Combine customers, orders, products and order_items.
Subqueries / CTEs	Compare results with averages and build reusable steps.
CASE WHEN	Create business categories such as order-value bands.
Window functions	Rank products or customers and calculate running totals.

7. Business Questions and SQL Patterns

What is the total revenue and average order value?

```
SELECT SUM(total_amount) AS total_revenue, AVG(total_amount) AS average_order_value FROM orders;
```

Which cities generate the most revenue?

```
SELECT c.city, SUM(o.total_amount) AS revenue FROM customers c JOIN orders o ON  
c.customer_id=o.customer_id GROUP BY c.city ORDER BY revenue DESC;
```

Which products generate the highest revenue?

```
SELECT p.product_name, SUM(oi.quantity*oi.unit_price) AS revenue FROM order_items oi JOIN products p ON  
oi.product_id=p.product_id GROUP BY p.product_name ORDER BY revenue DESC;
```

Which customers have the highest lifetime revenue?

```
SELECT c.customer_name, SUM(o.total_amount) AS revenue FROM customers c JOIN orders o ON  
c.customer_id=o.customer_id GROUP BY c.customer_id, c.customer_name ORDER BY revenue DESC;
```

What is the monthly revenue trend?

```
SELECT DATE_FORMAT(order_date, '%Y-%m') AS month, SUM(total_amount) AS revenue FROM orders GROUP BY  
month ORDER BY month;
```

Which products perform above the average product revenue?

```
WITH product_sales AS (...) SELECT * FROM product_sales WHERE revenue > (SELECT AVG(revenue) FROM  
product_sales);
```

8. Analytical Results

Metric	Result
Total revenue (order totals)	■2,944,268.00
Number of orders	150
Total units sold	1,132
Average order value	■19,628.45
Highest single order	■73,083.00
Number of cities	10

Revenue by City

City	Revenue
hyderabad	■368,077.00
bengaluru	■324,265.00
delhi	■322,961.00
chandigarh	■317,707.00
gurugram	■310,486.00
mumbai	■309,195.00
noida	■285,585.00
pune	■269,569.00
ahmedabad	■269,376.00
jaipur	■167,047.00

Revenue by Category

Category	Units sold	Revenue
electronics	466	■1,318,734.00
furniture	133	■738,067.00
fitness	192	■627,308.00
accessories	126	■169,974.00
stationery	159	■51,041.00
lifestyle	56	■39,144.00

Top Products

Product	Category	Units	Revenue
office chair	furniture	71	■638,929.00
running shoes	fitness	67	■334,933.00
smart watch	electronics	51	■305,949.00
headphones	electronics	67	■267,933.00
fitness band	fitness	67	■234,433.00
bluetooth speaker	electronics	67	■200,933.00
webcam	electronics	50	■164,950.00
power bank	electronics	72	■129,528.00
mechanical keyboard	electronics	44	■109,956.00

Product	Category	Units	Revenue
usb-c hub	electronics	68	■101,932.00

Monthly Revenue

Month	Revenue
2024-03	■410,765.00
2024-08	■351,474.00
2024-05	■332,763.00
2024-01	■299,294.00
2024-10	■282,874.00
2024-12	■249,705.00
2024-06	■230,311.00
2024-11	■228,627.00
2024-02	■172,827.00
2024-04	■146,446.00
2024-07	■121,631.00
2024-09	■117,551.00

9. Customer Analysis

Customer-level analysis helps identify high-value accounts and supports segmentation for retention campaigns, personalised recommendations and loyalty programmes. The following table lists the ten customers with the greatest recorded order revenue.

Customer	City	Orders	Revenue
navya bhatia 19	hyderabad	5	■146,056.00
rohan verma 25	chandigarh	6	■138,163.00
navya bhatia 39	hyderabad	4	■126,068.00
ishaan gupta 6	mumbai	5	■118,264.00
sara mehta 13	noida	5	■112,856.00
aarav sharma 21	delhi	3	■99,365.00
kiara verma 15	chandigarh	3	■94,772.00
myra gupta 36	mumbai	4	■94,274.00
reyansh khanna 30	ahmedabad	5	■92,856.00
vivaan kapoor 2	gurugram	4	■91,369.00

10. Key Findings

- The dataset records 150 orders and total order revenue of ■2,944,268.00.
- Hyderabad is the highest-revenue city, contributing ■368,077.00.
- Office Chair is the top product by line revenue at ■638,929.00.
- Electronics is the strongest category by line revenue.
- The highest monthly revenue occurs in 2024-03, with ■410,765.00.
- The largest recorded customer revenue belongs to Navya Bhatia 19 at ■146,056.00.

11. Recommendations

- Prioritise inventory and promotions for the leading products and categories.
- Create customer segments using order frequency, total revenue and average order value.
- Investigate low-revenue cities for targeted marketing or product assortment changes.
- Use monthly revenue trends to plan campaigns and stock levels.
- Add payment, delivery, discount and return fields in a future version for deeper profitability analysis.

12. Conclusion

This e-commerce SQL project demonstrates how a relational dataset can be transformed into useful business insights. By combining customer, order, product and order-item data, the analysis provides a complete view of sales performance. The project also demonstrates practical SQL skills, including relational joins, aggregation, ranking, CTEs and window-function-based analysis.